

JalRakshak - RTRWH / AR Assessment Report

Adyar, Chennai | Generated for demo purposes

Feasibility

Highly Feasible

Site Inputs

Roof area	120.0 sqm
Roof type	concrete
Open space	small_yard (10.0 sqm)

Rainfall & Soil

Annual rainfall	1238.0 mm (Chennai climatology fallback (Open-Meteo unreachable))
Soil texture	sandy (Chennai soil zone fallback table)
Infiltration rate	30.0 mm/hr
Nearest CGWB station	Adyar (0.0 km away)
Depth to water table	4.8 m (2025-05-01)
Elevation	6.0 m

Harvestable Volume

Deterministic estimate	126276.0 litres/year
Model range (P10 - P90)	101021.0 - 145217.0 litres/year
Model median (P50)	126276.0 litres/year
Model confidence	60.0%

Recommended Structure

Recharge Pit
Diameter M: 1.8 Depth M: 1.5

Shallow-to-moderate water table and a modest roof/open-space footprint make a standard recharge pit the most practical CGWB-recommended structure here.

Data Notes

- Rainfall used a climatology fallback because Open-Meteo was unreachable.
- Soil data used the hardcoded Chennai soil-zone fallback table, not live SoilGrids.
- Groundwater depth is from the nearest of 15 seeded CGWB monitoring stations (Adyar, 0.0km away), not a live India-WRIS query.
- Elevation used a flat Chennai-average fallback because OpenTopoData was unreachable.

This report is generated using publicly available rainfall (Open-Meteo), soil (SoilGrids / fallback zone table), groundwater (CGWB public monitoring data), and elevation (OpenTopoData) sources, combined with a CGWB-style deterministic sizing model and a supplementary ML estimate. It is intended as a preliminary planning aid and does not replace a site survey or an official CGWB assessment. © JalRakshak (DSC demo project).